There are regularities, represented by what we call physical laws, that link events of different times, but they are symmetric between future and past. In a microscopic description, there can be no sense in which the past is different from the future.*

This is the disconcerting conclusion that emerges from Boltzmann's work: the difference between the past and the future refers only to *our own* blurred vision of the world. It's a conclusion that leaves us flabbergasted: is it really possible that a perception so vivid, basic, existential—my perception of the passage of time—depends on the fact that I cannot apprehend the world in all of its minute detail? On a kind of distortion that's produced by myopia? Is it true that, if I could see exactly and take into consideration the actual dance of millions of molecules, then the future would be "just like" the past? Is it possible that I have as much knowledge of the past—or ignorance of it—as I do of the future? Even allowing for the fact that our perceptions of the world are frequently wrong, can the world really be *so* profoundly different from our perception of it as this?

All this undermines the very basis of our usual way of understanding time. It provokes incredulity, just as much as the discovery of the movement of the Earth did. But just as with the movement of the Earth, the evidence is overwhelming: all the phenomena that characterize the flowing of time are reduced to a "particular" state in the world's past, the "particularity" of which may be attributed to the blurring of our perspective.

Later on, I will delve into the mystery of this blurring, to see how it is tied to the strange initial improbability of the universe. For now, I will end with the mind-boggling fact that entropy, as Boltzmann fully understood, is nothing other than the number of microscopic states that our blurred vision of the world fails to distinguish.

The equation that states precisely this[20] is carved on Boltzmann's tomb in Vienna, above a marble bust that portrays him as an austere and surly figure, such as I don't believe he ever was in life. Many young students of physics go to visit his tomb, and linger there to ponder. And sometimes the odd elderly professor of physics as well.

Time has lost another of its crucial components: the intrinsic difference between past and future. Boltzmann understood that there is nothing intrinsic about the flowing of time. That it is only the blurred reflection of a mysterious improbability of the universe at a point in the past.

The source of Rilke's "eternal current" is nothing other than this.

Appointed a university professor at just twenty-five years old; received at court by the emperor at the apex of his success; severely criticized by the majority of the academic world, which did not understand his ideas; always